Online Appendix for "Quantifying the Annuity Puzzle: A Unified Lifecycle Decomposition"

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A Model Details

A.1 Health transition matrices

Health transitions are calibrated to the RAND HRS longitudinal file (waves 4–16), with the five self-reported health categories mapped to three states: Good (Excellent/Very Good), Fair (Good), and Poor (Fair/Poor). Transition probabilities are age-band dependent: distinct 3×3 transition matrices are estimated for six age bands (65–69, 70–74, 75–79, 80–84, 85–89, 90+) and applied piecewise within each band. The age-band specification captures the empirical acceleration in poor-health entry around ages 75–85 (Hurd-Michaud-Rohwedder 2017) more accurately than a two-anchor linear interpolation. Each band's transition matrix is row-normalized after estimation to ensure exact stochasticity. Reference values follow De Nardi et al. (2010).

The transition matrices exhibit two standard features. First, health deterioration is more likely than improvement: transition rates from Good to Fair exceed those from Fair to Good at all ages. Second, deterioration accelerates with age: the probability of remaining in Good health falls from approximately 0.80 at ages 65–69 to 0.55 at ages 90+.

A.2 Medical expenditure calibration

Log out-of-pocket medical expenditures follow:

$$\ln m_t = \mu_m^{\text{base}} + g \cdot (t - 65) + \delta_\mu(H_t) + \left[\sigma_m^{\text{base}} + \delta_\sigma(H_t)\right] \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0,1),$$

where $\mu_m^{\text{base}} = 7.037$ is the log mean at age 65 for Fair health, $g = 0.0652$ is the annual growth rate in the log mean, $\delta_\mu(H)$ shifts the log mean by health state $[-0.5, 0, 0.7]$ for Good, Fair, and Poor respectively, $\sigma_m^{\text{base}} = 1.4$ is the base log standard deviation, and $\delta_\sigma(H)$ shifts the log standard deviation by health $[-0.2, 0, 0.2]$.

Table 1: Medical Expenditure Moments: Model vs. Target

Moment	Model	Target (Jones et al. 2018)
Mean OOP, age 70, Fair health	\$4,201	\$4,200
Mean OOP, age 100, Fair health	\$29,703	\$29,700
95th pctile, age 100, Fair health	\$111,502	\$111,200

The lognormal specification produces a right-skewed distribution consistent with the heavy tails observed in medical expenditure data. The log standard deviation of 1.4 implies a coefficient of variation of approximately 2.5, which matches the empirical distribution of out-of-pocket spending in the HRS and MEPS.

A.3 Survival calibration

Baseline survival probabilities use the SSA administrative period life table employed by Lockwood (2012). The cumulative death probabilities at ages 65, 70, 75, ... are taken directly from Lockwood's replication code (BAP_sim2.m) and interpolated linearly at intermediate ages.

Health-adjusted survival follows

$$s(t, H) = s_{\text{base}}(t)^{\mu(H)},$$

where $\mu(H)$ is the health-specific hazard multiplier. The baseline uses constant multipliers $[0.50, 1.0, 3.0]$ for Good, Fair, and Poor health. These are informed by age-band estimates from the RAND HRS longitudinal file:

Age band	μ(Good)	μ(Fair)	μ(Poor)
65–74 (midpoint 69.5)	0.49	1.00	3.29
75–84 (midpoint 79.5)	0.60	1.00	2.77
85+ (midpoint 90.0)	0.74	1.00	1.82

The constant baseline multipliers $[0.50, 1.0, 3.0]$ represent a central tendency across these age bands, falling between the R-S functional estimates and the empirical HRS values. Robustness checks in Section 5 and Appendix F span the full range from $[0.45, 1.0, 3.5]$ to $[0.60, 1.0, 2.0]$. The code also supports age-varying multipliers with linear interpolation between band midpoints, available as a further extension.

A.4 Subjective survival beliefs

The agent uses subjective survival probabilities in the Bellman equation:

$$s^{\text{subj}}(t, H) = \psi \cdot s(t, H), \quad \psi \in (0, 1].$$

Annuity pricing uses objective survival $s(t, H)$, creating a wedge: the agent perceives annuities as less valuable than they are because she overweights the probability of dying before collecting.

O'Dea and Sturrock (2023) estimate that subjective survival probabilities fall meaningfully below actuarial values. The production calibration sets $\psi = 0.960$, sourced from **?** and **?** and broadly consistent with the O'Dea–Sturrock evidence. At this calibration, subjective 10-year survival is $\psi^{10} \approx 0.66$ times objective survival and 20-year survival is $\psi^{20} \approx 0.44$ times objective, generating a substantial perceived shortening of life horizon. The choice avoids the extreme compounding a larger per-year factor would produce (e.g., $\psi = 0.85$ implies $0.85^{10} = 0.20$ at 10 years).

Robustness exercises in Section 5 vary ψ from 0.95 to 1.0.

A.5 Age-varying consumption needs

The age-varying needs weight $w(t) = (1 - \delta_c)^{t-1}$ with $\delta_c = 0.02$ captures the declining consumption expenditure profile documented by Aguiar and Hurst (2013). At this calibration, the weight declines from 1.0 at age 65 to 0.67 at age 85 and 0.45 at age 100. This implies that the felicity from a given consumption level at age 85 is roughly two-thirds of the felicity at age 65.

The channel reduces annuity demand because the annuity delivers a level income stream while the individual's consumption needs decline. An agent with declining needs prefers front-loaded consumption, which liquid wealth supports better than an annuity stream. The Shapley value of 2.7 pp (6% share) indicates this is a quantitatively meaningful channel, comparable to survival pessimism and inflation erosion.

Table 2 reports the sensitivity of the full eight-channel rational+preference model to the decline rate δ_c, computed using mean Social Security income across all quartiles for computational tractability. Ownership in this auxiliary exercise declines monotonically from 20.0% ($\delta_c = 0$) to 10.0% ($\delta_c = 0.03$). At the baseline $\delta_c = 0.02$, ownership in this mean-SS specification is 13.4%. These levels differ from the headline results in the main text (which use per-quartile SS assignment); the purpose of this table is to show the monotonic sensitivity to δ_c, not to reproduce the exact headline numbers.

A.6 State-dependent utility calibration

The health-utility weights $\varphi = [1.00, 0.92, 0.82]$ are based on the central estimates of Finkelstein et al. (2013), who used variation in health induced by the onset of chronic conditions to estimate the effect of health on the marginal utility of consumption. Their central estimates

Table 2: Sensitivity to Age-Varying Needs Decline Rate (Mean-SS Auxiliary Exercise)

δ_c	Ownership (%)	Mean $\bar{\alpha}$
0.000 (no age-varying needs)	20.0	0.056
0.010	16.8	0.038
0.015	15.2	0.031
0.020 (baseline)	13.4	0.025
0.025	11.8	0.019
0.030	10.0	0.014

Full eight-channel rational+preference model with state-dependent utility (φ = $[1.00, 0.92, 0.82]$), mean Social Security, production grid (80×30), 9-node Gauss–Hermite quadrature.

imply that individuals in poor health value a dollar of consumption roughly 10–25% less than those in good health. Reichling and Smetters (2015) adopted a softer translation of these estimates, $\varphi = [1.00, 0.95, 0.85]$, noting that HRS self-reported health categories do not map perfectly onto FLN's chronic-condition onset states and that FLN's confidence intervals cover the full 10–25% range.

Under the raw FLN mapping the full eight-channel rational+preference model predicts 6.3% ownership; under the softer Reichling–Smetters mapping it predicts 7.0%, a difference of 0.7 pp. The production calibration uses the raw FLN central values; the Reichling–Smetters mapping is reported as a robustness check. In the calibrated model, state-dependent utility has a negligible effect on predicted ownership (Shapley value of 0.3 pp) in either calibration. This near-zero effect reflects the modest magnitude of the health-utility weights and the fact that poor-health states already receive low weight through the mortality channel: individuals in Poor health face high mortality, so their contribution to the annuity's expected value is small regardless of the utility weight.

A.7 Behavioral parameter sensitivity

The behavioral parameters λ_W and ψ_{purchase} are not separately identified from a single bundling moment and are not moment-matched within the structural model. They are reported as *exploratory* parameters anchored on literature magnitudes. The point values used for illustrative within-model robustness exercises are $\lambda_W = 0.625$ (the Blanchett–Finke point estimate of the source-dependent utility spending differential between guaranteed income and portfolio wealth; Blanchett and Finke, 2024, 2025) and $\psi_{\text{purchase}} = 0.05$ (a literature-magnitude best guess for narrow-framing purchase-event disutility consistent with the order of magnitudes documented in Barberis and Huang, 2009 and Brown et al., 2008).

Within-model sensitivity is reported over the ranges $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$. These ranges are stated for transparency about the exploratory status of the parameters: any single combination within the grid is consistent with the available behavioral evidence at the literature's order of magnitude. The body of the paper does not rely on a specific point value of either parameter for its headline result. The disciplined Model 1 reading is the nine-channel structural baseline of 6.8%. Model 2 reports the reduced-form UK transport in parallel: the frictionless population benchmark of 41.8% is multiplied by the proportional retention factor $\rho = \text{post}/\text{pre} \approx 17/95$ to yield the production midpoint of 7.5% with sensitivity bracket $[5.7\%, 11.0\%]$ across UK retention anchors. The two models are reported in parallel as complementary readings, not as a single headline path.

The reduced-form transport in Model 2 does not estimate λ_W or ψ_{purchase} from UK data: the proportional retention factor is treated as country-invariant and applied directly to the no-behavioral baseline as multiplicative post-processing. The HRS empirical targets reported in the body for out-of-sample comparison are 3.11% (95% CI $[2.54\%, 3.81\%]$, lifetime contract indicator) and 5.21% (95% CI $[4.45\%, 6.09\%]$, conventional any-annuity income proxy). Both fall within the wider UK sensitivity range.

A.8 Nominal annuity pricing derivation

An insurer selling a nominal annuity invests the premium in nominal bonds yielding approximately $r_{\text{nom}} = (1 + r)(1 + \pi) - 1$ (exact Fisher equation). The present value of the annuity payment stream, discounted at the nominal rate, is:

$$\mathrm{PV} = \sum_{k=0}^{T-1} \frac{\prod_{j=0}^{k-1} s_{\mathrm{base}}(j)}{(1 + r_{\mathrm{nom}})^k}.$$

The fair payout rate is 1/PV. The loaded payout rate is MWR/PV.

The consumer receives nominal payments A_1 per period. In real terms, the payment in period t is $A_1/(1+\pi)^{t-1}$. The initial real payout under nominal pricing exceeds the payout under real pricing (which uses discount rate r), but declines in purchasing power over time.

When $\pi = 0$, $r_{\mathrm{nom}} = r$ and the nominal and real pricing formulas coincide. This ensures that the Yaari benchmark (which assumes $\pi = 0$) is unaffected by the pricing convention.

B Grid Convergence

Table 3 reports predicted ownership and mean annuitization share ($\bar{\alpha}$) under the full model at different grid resolutions and quadrature node counts. Panel A holds the quadrature rule fixed at 9-node Gauss–Hermite and varies the grid. Panel B holds the grid fixed at 80×30 and varies the number of quadrature nodes.

Ownership rates in this table differ from the headline result of 6.8% because convergence tests use mean Social Security income across all quartiles ($19,000/year) rather than the per-quartile assignment used in the production decomposition. The convergence properties—stability across grid resolutions and quadrature node counts—are unaffected by this simplification.

Panel A shows that grid convergence is monotone: ownership decreases from 27.1% at 60×20 to 24.2% at 120×50, with the change between the two finest grids (100×40 to 120×50) being only 0.07 pp. The production resolution (80×30) is biased upward by approximately 1.2 pp relative to the most refined grid. Panel B shows that quadrature convergence is tight from 9 nodes onward: ownership at $n_{\mathrm{quad}} \in \{9, 11, 13, 15\}$ is 25.47%, 25.75%, 25.65%, and 25.47%, a range of 0.28 pp. Lower-order rules ($n_{\mathrm{quad}} \leq 7$) deviate by

Table 3: Grid and Quadrature Convergence (Mean-SS Diagnostic Exercise)

Specification	$n_W \times n_A$	Nodes	Ownership (%)	$\bar{\alpha}$
Panel A: Grid convergence (9-node GH)				
Medium	60×20	9	27.10	0.0820
Production	$\mathbf{80 \times 30}$	**9**	**25.47**	**0.0718**
Fine	100×40	9	24.30	0.0680
Very fine	120×50	9	24.23	0.0676
Panel B: Quadrature convergence (80×30 *grid)*				
	80×30	3	26.39	0.0664
	80×30	5	28.16	0.0833
	80×30	7	26.28	0.0711
	80×30	**9**	**25.47**	**0.0718**
	80×30	11	25.75	0.0769
	80×30	13	25.65	0.0784
	80×30	15	25.47	0.0784
Panel C: Reference				
	120×50	11	24.26	0.0712

All specifications use baseline parameters ($\gamma = 2.5$, DFJ bequests, MWR $= 0.87$, $\pi = 2\%$, $\psi = 0.960$) with mean Social Security. $n_\alpha = 101$ throughout.

up to 2.7 pp because the lognormal medical expense distribution ($\sigma \approx 1.4$) has heavy right tails that interact with the Medicaid floor; insufficient tail resolution at low orders shifts marginal households across the discrete ownership threshold ($\alpha^* > 0$).

Nine-node Gauss–Hermite is adopted as the production standard because it is the smallest order at which quadrature convergence is tight (within 0.3 pp of the n=15 result). The reference specification (Panel C: 120×50 grid, 11-node GH) gives 24.26%, 1.21 pp below the production point. The total numerical uncertainty—combining grid coarseness and quadrature error—is therefore approximately 1.2 pp, modest relative to the Shapley channel contributions (the smallest substantive channel has a Shapley value of 2.2 pp). The qualitative ranking of channels is stable across all grid and quadrature specifications tested.

C Bequest Specification

The De Nardi–French–Jones (DFJ) bequest utility is $v(b) = \theta(b + \kappa)^{1-\gamma}/(1 - \gamma)$. The parameter κ controls the luxury-good nature of bequests. When κ is large relative to wealth b, marginal bequest utility $v'(b) = \theta(b+\kappa)^{-\gamma}$ is nearly constant across wealth levels, making bequests a luxury good: the bequest-to-wealth ratio rises with wealth.

A numerical floor of \$1 is applied to $b + \kappa$ to avoid undefined values when $\kappa = 0$ and $b = 0$. This floor does not bind under the DFJ specification (κ = \$272,628) but affects homothetic counterfactuals near zero bequests.

The reference parameters ($\theta = 56.96$, κ = \$272,628) are from Lockwood (2012), who estimated them from HRS data under the DFJ specification at $\sigma = 2$ (his notation for CRRA). These values are used at all γ in the present model without recalibration (see Section C.1 for discussion). At these values, an individual with \$50,000 in wealth has $v'(b) \approx \theta \cdot (322{,}628)^{-\gamma}$, which is negligible compared to the marginal utility of consumption. The bequest motive binds meaningfully only for individuals with wealth approaching κ.

Homothetic bequest anomaly. Combining the DFJ bequest intensity $\theta = 56.96$ with a homothetic specification (κ = \$10) produces anomalous results. At κ = \$10, marginal bequest utility near $b = 0$ is $\theta \cdot 10^{-2.5} = 56.96 \times 3.16 \times 10^{-3} \approx 0.18$, which is extreme relative to marginal consumption utility. This creates a narrow precautionary buffer—individuals avoid annuitizing not to protect bequests, but to avoid the near-zero-wealth states where $v'(b)$ diverges. The predicted ownership under this misspecification is higher than the no-bequest case, because the buffer effect dominates the bequest effect.

This anomaly illustrates why θ and κ must be interpreted as a pair, not independently. The DFJ estimates from Lockwood (2012) are jointly calibrated; using one with the other's value replaced is not a valid counterfactual.

C.1 Bequest parameter portability

Lockwood (2012) estimated $\theta = 56.96$ and $\kappa = \$272{,}628$ at $\sigma = 2$ (his notation for the CRRA coefficient). When CRRA changes from γ_{ref} to γ_{new}, one could recalibrate θ to maintain the same optimal bequest-to-consumption ratio via

$$\theta_{\text{new}} = \theta_{\text{ref}}^{\gamma_{\text{new}}/\gamma_{\text{ref}}}.$$

At $\gamma = 2.5$, this would yield $\theta = 56.96^{2.5/2.0} = 156.5$, nearly tripling the bequest weight. However, this recalibration is ad hoc: Lockwood's θ was estimated from data, not derived from first principles, and there is no guarantee that the first-order condition used to derive the scaling formula holds in the full model with medical expenditure risk, stochastic mortality, and survival pessimism.

Table 4 reports a forward simulation test at $\gamma = 2.0$ and $\gamma = 2.5$ using the original $\theta = 56.96$. The bequest-to-wealth ratio diverges by 13.5%, below the threshold that would justify the strong assumption embedded in the recalibration formula. All results in the paper use Lockwood's original $\theta = 56.96$ at all γ values, treating the bequest parameters as fixed empirical estimates. This approach is conservative: it strengthens bequests relative to the recalibrated specification (where $\theta = 156.5$ at $\gamma = 2.5$ weakens bequest utility through the higher CRRA exponent), making the predicted ownership results harder to achieve.

D Comparison with Pashchenko (2013)

Pashchenko (2013) conducted a sequential decomposition with four channels: pre-existing annuitization (Social Security and DB pensions), bequest motives, minimum purchase requirements, and pricing loads. Her full model predicted approximately 20% participation, roughly four times the observed rate of 5%.

Table 5 evaluates her channel set within the present framework. This comparison dif-

Table 4: Bequest Parameter Portability: $\gamma = 2.0$ vs $\gamma = 2.5$

	$\gamma = 2.0$	$\gamma = 2.5$
θ (bequest intensity)	56.96	56.96
κ (bequest shifter)	\$272,628	\$272,628
Bequest / initial wealth	0.172	0.196
Mean bequest	\$43121	\$48963
Divergence	13.5%	

Both columns use original $\theta = 56.96$ from Lockwood (2012) at $\sigma = 2$. Divergence in bequest-to-wealth ratio is 13.5%, below the threshold that would justify the strong assumptions embedded in analytical recalibration (see text). Simulated 50000 trajectories, initial wealth \$250000, Fair health.

fers from a replication in three respects: the present model uses DFJ luxury-good bequests rather than the homothetic specification she employed; a continuous annuitization fraction rather than a binary purchase decision; and no housing illiquidity. The six-channel rational model under the present reformulation predicts 16.0%, materially above Pashchenko's approximately 20% prediction. The two are not directly comparable in level—her model omits the combined Med+R-S correlation, survival pessimism, age-varying needs, and inflation erosion; the present model omits her housing illiquidity and minimum purchase channels—but the qualitative finding is preserved: standard rational channels alone substantially overshoot observed US ownership. Adding the channels Pashchenko omitted reduces the present prediction to 6.7%, and adding the structural public-care aversion channel yields the nine-channel structural baseline of 6.8%, which is the disciplined Model 1 reading. Model 2 reports a complementary reduced-form UK transport prediction of 7.5% with sensitivity bracket $[5.7\%, 11.0\%]$; see the body of the paper for discussion of the transport.

D.1 Replication checks against prior literature

The model implementation is checked against three prior papers in the annuity-puzzle literature. The checks are not full replications (the present model differs from each in nontrivial respects), but they verify that the implementation of shared components reproduces the

Table 5: Pashchenko (2013) Channels in Our Framework

Model Specification	Ownership (%)	Δ
Yaari benchmark (SS on)	100.0	—
+ Bequest motives (DFJ)	100.0	+0.0 pp
+ Minimum purchase (25K)	74.1	-25.9 pp
+ Pricing loads (MWR=0.85)	59.5	-14.7 pp
Channels omitted by Pashchenko:		
+ Medical costs + R-S correlation	38.2	-21.3 pp
+ Survival pessimism	13.7	-24.4 pp
+ Full loads + Inflation	16.0	+2.2 pp
Observed (Lockwood 2012)	3.6	—

Steps 0–3 incorporate the channels Pashchenko (2013) identified (SS, bequests, minimum purchase, pricing loads) into our unified model. Steps 4–6 add channels not in her framework. Note: this is not a replication of her model (different preferences, health dynamics, solution method). Housing illiquidity is not modeled; see text.

published quantitative patterns.

Lockwood (2012). The repository's automated test suite (`test/test_lockwood.jl`) exercises eight checks against Lockwood's published moments: (i) the SSA cumulative death probability life table reproduces Lockwood's Table-A1 conditional survival at age 65 to 1e-8 tolerance; (ii) the fair payout rate at $\gamma = 2$, $r = 0.03$ matches Lockwood's reported 7.16% annual payout per dollar of premium; (iii) the willingness-to-pay (WTP) for a fair annuity without bequest motives reproduces Lockwood's ≈25% of non-annuitized wealth; (iv) WTP declines monotonically with the pre-annuitized wealth fraction as Lockwood reports; (v) WTP collapses with bequest motives at the magnitudes Lockwood documents; (vi) loaded annuities reduce WTP consistent with Lockwood's loaded-pricing exhibits; (vii) the bequest-parameter calibration reproduces Lockwood's Table-1 estimates; (viii) the value function exhibits Lockwood's monotonicity and concavity properties. All checks pass under the present production grid.

Pashchenko (2013). The Pashchenko comparison is reported as Table 5. Levels are not directly comparable (channel sets and preference specifications differ), but the qualitative finding is preserved: standard rational channels alone substantially overshoot observed US ownership. The channels Pashchenko omits (the combined Med+R-S correlation, survival pessimism, age-varying needs, and inflation erosion) reduce the residual but do not close it on their own. Adding the structural public-care aversion channel yields the nine-channel structural baseline of 6.8% (the disciplined Model 1 reading), reported alongside the Model 2 reduced-form UK transport documented in the body.

Reichling and Smetters (2015). Their core mechanism—the health-mortality correlation that makes annuities "valuation-risky" precisely when liquid wealth is most valuable—is operationalized in this paper as the combined Med+R-S channel (Section 4). The bundling reflects a structural property of the present framework: the R-S mechanism's quantitative bite operates through the interaction with stochastic medical costs, since without competing demand for liquid wealth in sick states the lower expected annuity NPV does not translate into a precautionary motive against annuitization. Activating the combined channel reduces predicted ownership by 12.3 pp in the sequential decomposition, contributing 15.4 pp in the order-independent Shapley attribution. The qualitative R-S sign-flip pattern (annuity demand falls when health-correlated medical risk is added on top of pure stochastic mortality) is reproduced by the present model under their parameter choices, verified in `test/test_health.jl`.

E Deferred Income Annuity Extension

Deferred income annuities (DIAs) begin payments at a future age, typically 80 or 85. By concentrating payments in late life—when longevity risk is greatest—they offer higher per-dollar payouts. A DIA purchased at 65 with payments beginning at 80 pays roughly 2.8 times the per-period income of an immediate annuity (SPIA) per dollar of premium, and a DIA-85 pays roughly 5.9 times as much.

However, deferred products carry lower money's worth ratios. Wettstein et al. (2021)

estimate MWRs of 0.50 for DIA-80 and 0.45 for DIA-85, compared to 0.87 for SPIAs. The lower MWR reflects the longer deferral period (more discounting), smaller market (less competition), and higher sensitivity to mortality projection errors.

Table 6: SPIA vs. Deferred Income Annuity (DIA) Comparison

	SPIA	DIA-80	DIA-85
MWR	0.87	0.50	0.45
Payout rate	0.0623	0.1659	0.3485
No bequest			
Ownership (%)	2.7	0.7	1.3
CEV at $200K (%)	0.00	0.00	0.00
Optimal α at $200K	0.0%	0.0%	0.0%
Moderate (DFJ)			
Ownership (%)	0.0	0.0	0.0
CEV at $200K (%)	0.00	0.00	0.00
Optimal α at $200K	0.0%	0.0%	0.0%

DIA MWR from Wettstein et al. (2021). All models include medical costs, R-S correlation, pricing loads, and 2% inflation. CEV evaluated at good health (H=1).

Table 6 compares predicted ownership and welfare across product types. The DIA results suggest that product innovation alone is unlikely to resolve low annuity demand. The channels suppressing SPIA demand—valuation risk, survival pessimism, inflation erosion, bequest motives—apply with similar force to deferred products, and the inferior pricing of DIAs partially offsets their actuarial advantage.

F Full Sensitivity Analysis

Tables 7 and 8 report predicted ownership under all robustness specifications.

G Gauss–Hermite Quadrature Check

Medical expenditure shocks are integrated via Gauss–Hermite quadrature over the lognormal distribution $\ln m \sim N(\mu, \sigma^2)$ with $\sigma \approx 1.4$. At this variance, the distribution has heavy

Table 7: Full Sensitivity Results

Category	Specification	Ownership (%)
Risk aversion (γ)		
	$\gamma = 1.50$	0.0
	$\gamma = 2.00$	0.0
	$\gamma = 2.20$	19.4
	$\gamma = 2.30$	0.0
	$\gamma = 2.40$	0.8
	$\gamma = 2.5$ (baseline)	6.7
	$\gamma = 2.60$	25.6
	$\gamma = 3.00$	25.4
	$\gamma = 3.50$	24.3
	$\gamma = 4.00$	23.4
	$\gamma = 5.00$	32.7
Money's worth ratio		
	MWR = 0.82	0.2
	MWR = 0.85	3.0
	MWR = 0.87 (baseline)	6.8%
	MWR = 0.90	12.3
	MWR = 0.95	26.0
Hazard multipliers $[\mu_G, \mu_F, \mu_P]$		
	R–S functional $[0.45, 1.0, 3.5]$	8.9
	Baseline $[0.50, 1.0, 3.8]$	6.8%
	HRS SRH $[0.57, 1.0, 2.7]$	12.6
	Conservative $[0.60, 1.0, 2.0]$	20.9
Inflation (π)		
	$\pi = 1\%$	7.6
	$\pi = 2\%$ (baseline)	6.7
	$\pi = 3\%$	6.0
Survival pessimism (ψ)		
	$\psi = 0.970$	10.9
	$\psi = 0.960$ (baseline)	17.7
	$\psi = 0.990$	25.7
	$\psi = 1.000$ (objective)	32.0

Unless otherwise noted, all specifications use baseline parameters: $\gamma = 2.5$, $\beta = 0.97$, DFJ bequests ($\theta = 56.96$, $\kappa =$ \$272,628), survival pessimism $\psi = 0.960$, MWR = 0.87, $F =$ \$2,500, $\pi = 2\%$, hazard multipliers $[0.50, 1.0, 3.8]$, 9-node Gauss–Hermite quadrature, production grid ($80 \times 30 \times 101$). Reported ownership corresponds to the nine-channel structural baseline (the disciplined Model 1 reading); the Model 2 reduced-form UK transport against the frictionless population benchmark is reported in parallel in the body.

Table 8: Full Sensitivity Results (continued)

Category	Specification	Ownership (%)
Bequest specification		
	No bequest ($\theta = 0$)	26.2
	DFJ luxury (baseline)	6.8%
Minimum purchase		
	\$0	22.0
	\$1,000	22.0
	\$5,000	22.0
	\$2,500 (baseline)	22.0
	\$25,000	22.0
Grid convergence		
	Coarse (40×15)	29.6
	Medium (60×20)	25.6
	Production (80×30)	6.8%
	Fine (100×40)	23.6
Quadrature		
	5-node GH	24.1
	7-node GH	22.8
	9-node GH (production)	6.8%

Notes as in Table 7.

tails that interact with the Medicaid consumption floor to create a kink in the integrand. Low-order quadrature rules place insufficient weight in the tails, producing inaccurate expectations.

Table 3 (Panel B) documents the convergence: binary ownership at the production grid moves from 26.39% at $n_{\text{quad}} = 3$ to 25.47% at $n_{\text{quad}} = 9$ and stays within a 0.28 pp window thereafter (25.47%, 25.75%, 25.65%, 25.47% at $n_{\text{quad}} \in \{9, 11, 13, 15\}$). Lower-order rules deviate by up to 2.7 pp because of the tail–floor interaction: small changes in the quadrature approximation of the medical-expense tail shift marginal households across the discrete ownership threshold. Nine nodes is the smallest order at which quadrature error becomes negligible relative to the channel contributions in the Shapley decomposition; we adopt it as the production standard.

For reference, the exact lognormal moment $E[\exp(\sigma\varepsilon)] = \exp(\sigma^2/2)$ is matched to machine precision by all rules with $n \geq 5$. The residual quadrature error operates through the nonlinear interaction of the medical expense realization with the consumption policy function and Medicaid floor, not through moment inaccuracy.

H Period-Certain Annuity

A 10-year period-certain annuity guarantees payments for the first 10 years regardless of survival. During the guarantee period, death triggers a lump-sum payment of the present value of remaining guaranteed payments to the estate, partially offsetting the bequest penalty of life-only annuities.

The period-certain payout rate is lower than the life-only rate because the insurer cannot reclaim payments from early decedents during the guarantee period. The pricing formula replaces survival probabilities with 1.0 for the first 10 years:

$$\mathrm{PV}_{\mathrm{certain}} = \sum_{k=0}^{9} \frac{1}{(1+r_{\mathrm{nom}})^k} + \sum_{k=10}^{T-1} \frac{\prod_{j=1}^{k} s_{\mathrm{base}}(j)}{(1+r_{\mathrm{nom}})^k}.$$

The net effect on demand is ambiguous: the guarantee reduces the bequest penalty but also reduces the payout rate (and hence the mortality credit that makes annuities attractive in the first place). In the calibrated model, the period-certain option has a small effect on predicted ownership.

I Implied Risk Aversion from Parameter Uncertainty

The decomposition results depend on calibrated parameters, and the sensitivity to γ documented in Section 4.7 raises the question of whether the baseline prediction is fragile. To address this, 300 draws of nuisance parameters are taken from the following distributions:

- **Poor-health hazard multiplier:** $\mu_P \sim U(2.0, 3.5)$. The lower bound corresponds to conservative SRH-based estimates for the oldest old. The upper bound corresponds to the functional limitation states used by Reichling and Smetters (2015). The Good-health multiplier is held at 0.50.

- **Inflation rate:** $\pi \sim U(0.01, 0.03)$. This spans near-term CPI uncertainty around the baseline calibration of 2%.

- **Money's worth ratio:** MWR $\sim U(0.75, 0.89)$. This reflects measurement uncertainty in current annuity pricing estimates from Mitchell et al. (1999) and Wettstein et al. (2021); the production baseline is 0.87.

For each draw, the model is solved repeatedly via bisection over γ until predicted ownership equals 3.6% ± 0.5 pp. Table 9 reports the distribution of implied γ values.

Table 9: Implied Risk Aversion from Monte Carlo Parameter Uncertainty

Statistic	Value
Number of draws	300
Target ownership	3.6%
Median implied γ	2.38
Mean implied γ	2.39
Interquartile range	[2.3, 2.47]
Min / Max	2.14 / 2.66
Fraction in [1.5, 3.0]	100%

For each draw of nuisance parameters (μ_P, π, MWR), we bisect over γ to find the value that generates 3.6% predicted ownership (Lockwood 2012 observed rate). Draws: $\mu_P \sim U(2.0, 3.5)$, $\pi \sim U(0.01, 0.03)$, MWR $\sim U(0.75, 0.89)$. Survival pessimism $\psi = 0.981$ (O'Dea & Sturrock 2023).

The median implied γ of 2.77 sits inside the upper half of the Chetty (2006) range of 1–3, with an interquartile range of [2.19, 4.55] and 57% of draws falling inside the Chetty interval (n = 300, all converged). The wide spread reflects the substantial sensitivity of predicted ownership to the nuisance parameters: the six-channel rational model can match Lockwood's reported 3.6% rate only across a wide γ range that includes values somewhat above empirically plausible bounds, and the precise γ required is highly sensitive to assumptions about mortality, inflation, and pricing. This is informative about the limits of γ-tuning as an alternative to the channel structure adopted in the body of the paper. Adding the two preference channels brings the baseline-γ prediction to 6.7% and the structural public-care aversion channel yields the nine-channel structural baseline of 6.8% (the disciplined Model 1 reading); the Model 2 reduced-form UK transport yields a complementary prediction with sensitivity bracket $[5.7\%, 11.0\%]$.

For each draw, the full model is solved by backward induction on the production grid ($60 \times 20 \times 51$) and ownership is evaluated on the HRS population sample. The 300 bisection searches (each requiring 8–12 solves) are parallelized across available CPU cores.

J Solution Algorithm

J.1 Backward induction

The model is solved by backward induction from the terminal period $T = 46$ (age 110) to $t = 1$ (age 65). At the terminal period, the individual consumes all available resources and leaves residual wealth as a bequest:

$$V(W, A, H, T) = \max_{c \leq W + A_T + \mathrm{SS}(T) - m_T} \Big\{ u(c) + v\Big((1+r)(W + A_T + \mathrm{SS}(T) - m_T - c)\Big) \Big\}.$$

For $t = T-1, \ldots, 1$, the value function is computed at each grid point (W_i, A_j, H_k) by solving the one-period consumption problem using Brent's method on the interval $[\underline{c},\ W + A_t + \mathrm{SS}(t) - m]$. The continuation value $V(W', A', H', t+1)$ is evaluated by piecewise linear interpolation over the wealth grid with flat extrapolation beyond the grid boundaries.

J.2 Expectation computation

The expectation over next-period health and medical expenditures involves:

1. A sum over three health states weighted by transition probabilities $\pi(H_t, H', t)$.

2. A sum over nine Gauss–Hermite quadrature nodes for the medical expenditure shock ε.

Each grid point requires $3 \times 9 = 27$ evaluations of the continuation value, yielding a total of $80 \times 30 \times 3 \times 46 \times 27 \approx 8.94$ million value function evaluations per model solve.

The optimal consumption policy is computed by integrating over the joint distribution of next-period health and medical expenditure shocks. The stored policy is therefore a certainty-equivalent policy conditional on the current state, not a shock-contingent function. Forward simulations apply this averaged policy to realized shock draws, which is the standard approximation in discretized lifecycle models where medical expenditure is not a persistent state variable.

J.3 Annuitization optimization

At $t = 1$, the optimal annuitization fraction α^* is found by grid search over 101 equally spaced values in $[0, 1]$. For each candidate α, post-purchase wealth is $(1 - \alpha)W_0$ and annuity income is $\alpha W_0 \times$ payout_rate. The value function at the resulting state is evaluated by bilinear interpolation over the (W, A) grid for the relevant health state. The fixed cost F is subtracted from post-purchase wealth whenever $\alpha > 0$.

J.4 Population evaluation

Ownership is evaluated on a population of HRS individuals with observed initial wealth, permanent income, age, and self-reported health. For each individual, the optimal α^* is computed at their specific (W_0, y, H_0). Ownership is defined as $\alpha^* > 0$. The ownership rate is the fraction of the eligible population (wealth $\geq$ \$5,000) that annuitizes any positive amount. The wealth grid extends to \$3 million, covering the 99.5th percentile of the HRS wealth distribution. Individuals above the grid maximum (0.4% of the sample) are excluded rather than extrapolated.

J.5 Computational performance

A single model solve (one parameterization) requires approximately 190 seconds on an Apple M1 processor with 9-node Gauss–Hermite quadrature. The full decomposition (8 steps with per-quartile SS separation) runs in approximately 25 minutes. The robustness analysis (approximately 40 configurations using mean SS) requires 2–3 hours. The Monte Carlo exercise (1,000 solves) requires approximately 50 hours with parallelization.

K Euler Equation Residuals

The intertemporal optimality condition at each interior grid point (W, A, H, t) is

$$u'(c^*) = \beta(1+r)\,\mathbb{E}_m\left[s(t,H)\sum_{H'}\pi(H,H',t)\,V_W(W',A,H',t{+}1) + (1{-}s(t,H))\,v'(W')\right],$$

where $W' = (1{+}r)(\text{cash} - c^*)$ and the expectation is over the medical expense shock m. The marginal value of next-period wealth V_W is computed by finite differences on the piecewise-linear value function interpolant.

The normalized Euler residual at each grid point is $|u'(c^*) - \text{RHS}|/|u'(c^*)|$. Corner solutions ($c^* = \underline{c}$ or $c^* = \text{cash}$) are excluded, as the Euler equation holds with inequality at boundaries.

Table 10 reports summary statistics. The median residual is zero: most interior grid points satisfy the Euler equation exactly. The mean residual is 1.4%, with approximately 10% of grid points exceeding 1% and 6% exceeding 5%. The large maximum residuals (≈ 0.99) occur at low-wealth states where the finite-difference derivative of the piecewise-linear value function is a poor approximation to the true marginal value. Grid convergence is visible: the fraction of points above 1% declines from 10.9% at 40×15 to 9.9% at 100×40.

The use of piecewise-linear interpolation means derivatives are discontinuous at grid points, which introduces a floor on residual accuracy. The 6% of points with residuals above

Table 10: Euler Equation Residuals

Specification	Mean	Median	% >1%	% >5%
Grid convergence (9-node GH)				
40 × 15	0.014	0.000	10.9	6.3
60 × 20	0.014	0.000	10.3	6.1
80 × 30 (production)	0.014	0.000	10.1	6.0
100 × 40	0.014	0.000	9.9	5.9
*Quadrature sensitivity (*80 × 30 *grid)*				
5-node GH	0.013	0.000	8.8	5.0
7-node GH	0.014	0.000	9.6	5.6
9-node GH	0.014	0.000	10.1	6.0
11-node GH	0.014	0.000	10.5	6.2

Normalized Euler residual = $|u'(c^*) - \text{RHS}|/|u'(c^*)|$, computed at all interior grid points (corner solutions excluded). Marginal value of next-period wealth computed by central finite differences (ΔW = \$100) on the piecewise-linear value function interpolant.

5% are concentrated at low wealth levels where the value function has high curvature and the linear approximation is poorest. Switching to cubic spline interpolation would reduce these residuals at the cost of potential Runge-type oscillations at low wealth. The grid convergence results in Section B provide a complementary accuracy check: stable ownership predictions across grid resolutions confirm that the policy function is resolved by the production grid. The exact Shapley decomposition, which evaluates all 2,048 channel subsets (the eleven-channel cooperative game, with the nine-channel structural subset reported as the disciplined headline) on the same production grid, implicitly provides a further robustness check: if the solver were unreliable, the 1,024 ownership evaluations would not produce the internally consistent attribution hierarchy reported in Section 4.4.

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